

EXCEPTION HANDLING(6/8/26)

1.

Q1 Handling ArithmeticException

10 marks

Write a Java program to divide two integers entered by the user. If the denominator is zero, handle the exception using a try-catch block and display an appropriate error message.

Your Code

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String [] args){
4         Scanner sc = new Scanner(System.in);
5         int a = sc.nextInt();
6         int b = sc.nextInt();
7         try{
8             int result = a/b;
9             System.out.println(result);
10        }catch(ArithmeticException e){
11            System.out.println("ArithmeticException: Cannot divide by zero");
12        }
13        sc.close();
14    }
15 }
```

Input

10
0

Output

ArithmeticException: Cannot
divide by zero

Run

Save

2.

Q2 Handling ArrayIndexOutOfBoundsException

10 marks

Write a Java program to create an integer array of five elements. Ask the user to enter an index and display the corresponding element. If the index is outside the valid range, handle the exception and display an error message.

Your Code

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String [] args){
4         Scanner sc = new Scanner(System.in);
5         int [] arr = new int[5];
6         for (int i = 0;i<5;i++){
7             arr[i] = sc.nextInt();
8         }
9         int index = sc.nextInt();
10        try{
11            System.out.println(arr[index]);
12        }catch(ArrayIndexOutOfBoundsException e){
13            System.out.println("ArrayIndexOutOfBoundsException");
14        }
15        sc.close();
16    }
17 }
```

Input

```
10 20 30 40 50
10
```

Output

```
ArrayIndexOutOfBoundsException: Invalid index
```

Run

Save

3.

Q3 Handling NumberFormatException

10 marks

Write a Java program to accept a number as a string and convert it into an integer using `Integer.parseInt()`. If the input is not a valid integer, handle the exception and display an appropriate message.

Your Code

```
1 import java.util.Scanner;  
2 public class Main{  
3     public static void main(String [] args){  
4         Scanner sc = new Scanner(System.in);  
5         String s = sc.nextLine();  
6         try{  
7             int n = Integer.parseInt(s);  
8             System.out.println(n);  
9         }catch(NumberFormatException e){  
10             System.out.println("NumberFormatException: In  
11         }  
12         sc.close();  
13     }  
14 }
```

 Run

 Save

Input

abc

Output

```
NumberFormatException:  
Invalid intger
```

4.

Q4 Handling Multiple Exceptions

10 marks

Write a Java program that performs the following operations:

Accept two integers from the user.

Divide the first number by the second.

Store the result in an array.

Handle both `ArithmeticException` and `ArrayIndexOutOfBoundsException` using multiple catch blocks.

Your Code

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String []args){
4         Scanner sc = new Scanner(System.in);
5         int[]arr = new int[1];
6         try{
7             int a = sc.nextInt();
8             int b = sc.nextInt();
9             int result = a/b;
10            arr[0]=result;
11            System.out.println(arr[0]);
12        }
13        catch(ArithmeticException e){
14            System.out.println("ArithmeticException: Cannot divide by
15        }catch(ArrayIndexOutOfBoundsException e){
16            System.out.println("ArrayIndexOutOfBoundsException: Invalid a
17        }
18        sc.close();
19    }
20 }
```

Input

10
0

Output

ArithmeticException: Cannot divide
by zero